

# Statistical and Machine Learning

## Tree based methods and boosting

### Instructions

All questions should first be discussed in pairs or small groups. Afterwards, we shall discuss them together. Solutions will be distributed after each session.

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### Exercise 1 — Decision Tree Terminology

Consider a decision tree fitted to predict baseball players' salaries using two predictors: **Years** and **Hits**. The tree produces three terminal regions:

$$R_1 = \{X \mid \text{Years} < 4.5\}, \quad R_2 = \{X \mid \text{Years} \geq 4.5, \text{ Hits} < 117.5\}, \quad R_3 = \{X \mid \text{Years} \geq 4.5, \text{ Hits} \geq 117.5\}.$$

- (a) Identify the **root node**, one **internal node**, and the **terminal nodes** (leaves).
- (b) A new player has 6 years of experience and 130 hits. Which region does this player fall into? If the average log-salary in that region is  $\hat{c}_m = 6.74$ , what is the predicted salary?
- (c) Explain the difference between how a **regression tree** and a **classification tree** make predictions in a terminal node.

### Solution

- (a)
    - **Root node**: the very first split,  $\text{Years} < 4.5$ .
    - **Internal node**: the node  $\text{Hits} < 117.5$ , only reached when  $\text{Years} \geq 4.5$ .
    - **Terminal nodes (leaves)**:  $R_1, R_2, R_3$  — the final regions where no further splitting occurs.
  - (b)

The player has  $\text{Years} = 6 \geq 4.5$  and  $\text{Hits} = 130 \geq 117.5$ , so they fall into  $R_3$ . The predicted log-salary is  $\hat{f}(x) = 6.74$ .
  - (c)
    - **Regression tree**: predicts the **mean response**  $\hat{c}_m = \frac{1}{|R_m|} \sum_{i: x_i \in R_m} y_i$  for all observations in the leaf.
    - **Classification tree**: computes class proportions  $\hat{\pi}_{mk}$  and predicts the **majority class**  $\hat{f}(x) = \arg \max_k \hat{\pi}_{mk}$ , or outputs class probabilities.
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### Exercise 2 — CART Split Selection

Suppose you are fitting a **regression tree** and considering two candidate splits on a node with 6 observations: responses  $y = (1, 3, 3, 5, 8, 10)$  and predictor values  $X = (2, 4, 5, 7, 9, 11)$ .

The two candidate splits are  $X < 5.5$  and  $X < 8$ .

(a) For each candidate split, identify which observations fall into  $R_1$  (left) and  $R_2$  (right), and compute the region means  $\hat{c}_1$  and  $\hat{c}_2$ .

(b) Compute the RSS for each candidate split:

$$\text{RSS}(j, s) = \sum_{i: x_i < s} (y_i - \hat{c}_1)^2 + \sum_{i: x_i \geq s} (y_i - \hat{c}_2)^2.$$

(c) Which split does CART select, and why?

(d) Does CART evaluate every possible real-valued split point  $s$ ? Explain how the search is made computationally feasible.

### Solution

(a)

- $X < 5.5$ :  $R_1 = \{1, 3, 3\}$ ,  $\hat{c}_1 = 7/3 \approx 2.33$ ;  $R_2 = \{5, 8, 10\}$ ,  $\hat{c}_2 = 23/3 \approx 7.67$ .
- $X < 8$ :  $R_1 = \{1, 3, 3, 5\}$ ,  $\hat{c}_1 = 3$ ;  $R_2 = \{8, 10\}$ ,  $\hat{c}_2 = 9$ .

(b)

- $X < 5.5$ :  $\text{RSS} = (1 - 2.33)^2 + (3 - 2.33)^2 + (3 - 2.33)^2 + (5 - 7.67)^2 + (8 - 7.67)^2 + (10 - 7.67)^2 \approx 15.34$ .
- $X < 8$ :  $\text{RSS} = (1 - 3)^2 + (3 - 3)^2 + (3 - 3)^2 + (5 - 3)^2 + (8 - 9)^2 + (10 - 9)^2 = 4 + 0 + 0 + 4 + 1 + 1 = 10$ .

(c)

CART selects  $X < 8$  because it gives the smaller RSS ( $10 < 15.34$ ), meaning the better reduction in prediction error.

(d)

No. CART sorts the observed values of  $X_j$  and evaluates only the **midpoints between consecutive distinct observed values** as candidate splits. Running sums of  $y_i$  and  $y_i^2$  allow efficient updating of the RSS as the split point moves, so no model is refitted from scratch for each candidate.

### Exercise 3 — Impurity Measures for Classification Trees

Consider a binary classification problem ( $K = 2$ ). A candidate split produces:

- **Left node**  $R_1$ : 40 observations — 30 from class 1, 10 from class 0.
- **Right node**  $R_2$ : 60 observations — 10 from class 1, 50 from class 0.

(a) Compute the class proportion  $\hat{\pi}_{m1}$  (proportion of class 1) in each node.

(b) For each node, compute the three impurity measures:

- Misclassification error:  $i_1(R_m) = 1 - \max_k \hat{\pi}_{mk}$
- Gini index:  $i_2(R_m) = 1 - \sum_k \hat{\pi}_{mk}^2$
- Entropy:  $i_3(R_m) = - \sum_k \hat{\pi}_{mk} \log(\hat{\pi}_{mk})$

(c) Compute the **weighted Gini impurity**  $i_w = \frac{|R_1|}{|R|} i_2(R_1) + \frac{|R_2|}{|R|} i_2(R_2)$ , where  $|R| = 100$ .

(d) Why is misclassification error often a less useful criterion for guiding splits than the Gini index or entropy?

### Solution

(a)

- $R_1$ :  $\hat{\pi}_{11} = 30/40 = 0.75$ ,  $\hat{\pi}_{10} = 0.25$ .
- $R_2$ :  $\hat{\pi}_{21} = 10/60 \approx 0.167$ ,  $\hat{\pi}_{20} \approx 0.833$ .

(b)

$R_1$ :  $i_1 = 1 - 0.75 = 0.25$ ;  $i_2 = 1 - (0.75^2 + 0.25^2) = 0.375$ ;  $i_3 \approx -(0.75 \log 0.75 + 0.25 \log 0.25) \approx 0.562$ .

$R_2$ :  $i_1 = 1 - 0.833 = 0.167$ ;  $i_2 \approx 1 - (0.833^2 + 0.167^2) \approx 0.278$ ;  $i_3 \approx 0.451$ .

(c)

$$i_w = \frac{40}{100}(0.375) + \frac{60}{100}(0.278) = 0.15 + 0.167 = 0.317.$$

(d)

The misclassification error is insensitive to changes in class proportions that do not alter the majority class — two quite different distributions may yield the same  $i_1$ . The Gini index and entropy are more sensitive to changes in  $\hat{\pi}_{mk}$  and provide a smoother, more informative signal for choosing among candidate splits.

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## Exercise 4 — Bagging and Out-of-Bag Error

(a) Describe in your own words the two main steps of the **bagging** algorithm: the bootstrap step and the aggregating step.

(b) A training dataset has  $m = 200$  observations. A bootstrap sample is drawn by sampling  $m$  observations **with replacement**.

- What is the probability that a given observation is **not** selected in a single draw?
- What is the approximate probability that a given observation does **not** appear in the bootstrap sample at all?
- Approximately how many observations are expected to be **out-of-bag (OOB)**?

(c) Explain how OOB observations can estimate the **test error** without a separate validation set.

(d) In bagging, all  $M$  trees receive equal weight  $\beta_j = 1/M$ . How does this differ from boosting?

### Solution

(a)

- **Bootstrap**: draw  $M$  samples of size  $m$  from  $\mathcal{D}$  with replacement. Some observations appear multiple times; others not at all.
- **Aggregating**: train one tree per bootstrap sample and combine predictions by averaging (regression) or majority vote (classification).

(b)

- Probability of not being selected in one draw:  $1 - 1/200 = 0.995$ .
- Probability of being absent from all  $m$  draws:  $(1 - 1/200)^{200} \approx e^{-1} \approx 0.368$ .
- Expected OOB observations:  $0.368 \times 200 \approx 74$ .

(c)

Each observation is OOB for roughly 37% of trees (those trained without it). Predicting it using only those trees and comparing to the true label gives an unbiased test-error estimate requiring no additional held-out data.

(d)

In bagging all trees are weighted equally. In boosting, each weak learner receives a weight  $\alpha_m$  (or is scaled by learning rate  $\nu$ ) reflecting its contribution to reducing the current loss — better-performing learners generally receive more influence.

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## Exercise 5 — Random Forests vs. Bagging

- (a) Describe the key algorithmic difference between **random forests** and standard bagging with trees. Why is this modification introduced?
- (b) For a classification problem with  $p = 100$  predictors, what is the typical recommended value of **mtry**?
- (c) If  $X_1, \dots, X_M$  are identically distributed with variance  $\sigma^2$  and pairwise correlation  $\rho$ , then

$$\text{Var}\left(\frac{1}{M} \sum_{m=1}^M X_m\right) = \rho\sigma^2 + \frac{1-\rho}{M}\sigma^2.$$

Use this to explain intuitively why **decorrelating trees** reduces ensemble variance.

- (d) In an ensemble of  $M = 3$  independent classifiers each correct with probability  $p = 0.6$ , compute the probability that the **majority vote** is correct. What happens if all classifiers are perfectly correlated?

### Solution

(a)

Standard bagging considers all  $p$  predictors at every split. Random forests restrict each split to a **random subset of mtry predictors**. This decorrelates the trees so that the ensemble variance is lower — strong predictors do not dominate every split in every tree.

(b)

$$\text{mtry} \approx \sqrt{p} = \sqrt{100} = 10.$$

(c)

The term  $\rho\sigma^2$  does not shrink as  $M$  grows — it reflects irreducible variance from correlated errors. Only the second term  $(1 - \rho)\sigma^2/M$  vanishes for large  $M$ . Reducing  $\rho$  (decorrelating the trees) makes the first term smaller, substantially lowering ensemble variance.

(d)

$$P = \binom{3}{2} 0.6^2 (0.4) + \binom{3}{3} 0.6^3 = 3(0.144) + 0.216 = 0.648 > 0.6.$$

With perfect correlation, all classifiers always agree, so the ensemble equals a single classifier — majority voting provides no improvement.

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## Exercise 6 — Boosting: Residuals, AdaBoost, and Gradient Boosting

- (a) In **boosting for regression trees**, successive trees are fitted to the **residuals** of the current ensemble. Explain the intuition.
- (b) Boosting has three key tuning parameters: the number of trees  $B$ , the shrinkage parameter  $\lambda$ , and the number of splits  $d$  per tree. Describe the role and trade-offs of each.
- (c) In **AdaBoost**, the weight assigned to classifier  $m$  is  $\alpha_m = \log\left(\frac{1 - \text{err}_m}{\text{err}_m}\right)$ . Compute  $\alpha_1, \alpha_2, \alpha_3$  for  $\text{err}_1 = 0.1$ ,  $\text{err}_2 = 0.3$ ,  $\text{err}_3 = 0.45$  and interpret the results.
- (d) Explain the analogy between **gradient boosting** and gradient descent. What are pseudo-residuals and how are they used?
- (e) XGBoost adds the regularization term  $\Omega(f) = \gamma J + \frac{1}{2}\lambda \sum_{j=1}^J w_j^2$ . Explain the role of  $\gamma$ ,  $\lambda$ ,  $J$ , and  $w_j$ .

## Solution

(a)

After  $b - 1$  trees, the residuals  $r_i = y_i - \hat{f}(x_i)$  capture what remains unexplained. Fitting the next tree to these residuals trains it to correct the current model's errors in regions where it performs poorly. The shrinkage  $\lambda$  prevents any single tree from over-correcting.

(b)

- $B$ : number of trees; larger  $B$  increases model capacity but can overfit if  $\lambda$  is too large.
- $\lambda$ : shrinkage; smaller values mean each tree contributes less, requiring a larger  $B$  but yielding better regularization. Typical values: 0.001–0.01.
- $d$ : tree depth;  $d = 1$  gives stumps with no interactions, larger  $d$  captures interactions between predictors but increases variance.

(c)

$$\alpha_1 = \log(0.9/0.1) = \log 9 \approx 2.20, \quad \alpha_2 = \log(0.7/0.3) \approx 0.85, \quad \alpha_3 = \log(0.55/0.45) \approx 0.20.$$

The most accurate classifier ( $\text{err}_1 = 0.1$ ) receives the highest weight. Classifier 3, barely better than random, contributes very little. If  $\text{err}_m = 0.5$ , then  $\alpha_m = 0$  and the classifier is ignored entirely.

(d)

Gradient descent updates parameters in the direction of the negative gradient of the loss. In gradient boosting, pseudo-residuals are the negative gradients of the loss with respect to the current predictions:

$$r_{im} = - \left[ \frac{\partial L(y_i, f(x_i))}{\partial f(x_i)} \right]_{f=f_{m-1}}.$$

A new weak learner  $F_m$  is fitted to these pseudo-residuals, and the model is updated as  $f_m(x) = f_{m-1}(x) + \nu F_m(x)$ , analogous to a gradient descent step with step size  $\nu$ .

(e)

- $J$ : number of leaves. The term  $\gamma J$  penalises trees with many leaves, discouraging overly complex structures (larger  $\gamma \Rightarrow$  fewer leaves).
- $w_j$ : prediction value in leaf  $j$ . The term  $\frac{1}{2}\lambda \sum_j w_j^2$  is an  $L_2$  penalty on leaf weights, shrinking them toward zero, analogous to ridge regularization.